

Chess Reinforcement Learning

Comparative Evaluation of Deep RL Algorithms

Environment: Custom Gymnasium Chess | Framework: Stable-Baselines3

1. Abstract

This report presents the evaluation results of six deep reinforcement learning algorithms applied to the game of chess. The algorithms evaluated are Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), Deep Q-Network (DQN), Double DQN (QR-DQN), Deep Deterministic Policy Gradient (DDPG), and Soft Actor-Critic (SAC). Each model was trained using a custom Gymnasium-compatible chess environment with a discrete action space of 4096 and a 14-channel (8x8) observation tensor. Models were benchmarked against a heuristic opponent over 20 evaluation episodes per algorithm.

3. Algorithm Summary

- PPO:** On-policy, clipped surrogate objective. Uses MaskablePPO with action masking for legal moves.
- A2C:** Synchronous advantage actor-critic. Simpler than PPO but faster per update.
- DQN:** Off-policy value-based with experience replay and target network.

2. Environment Design

The chess environment (ChessEnv) is built on python-chess and Gymnasium. Key specifications:

- Action Space: Discrete(4096) encoding from_sq*64+to_sq
- Observation: (14, 8, 8) float32 tensor with 12 piece planes, turn plane, and castling/check meta-plane
- Reward Shaping: Win (+10), Loss (-10), Material delta, center control (+0.05), check bonus (+0.2), step penalty (-0.01), illegal move (-1)
- Modes: Human-vs-AI, AI-vs-AI, Human-vs-Human
- Opponent: Heuristic bot with material-based evaluation

- DDQN:** Quantile Regression DQN (QR-DQN). Distributional RL reducing overestimation.
- DDPG:** Off-policy actor-critic for continuous actions. Uses discrete-to-continuous wrapper.
- SAC:** Maximum entropy RL with automatic temperature. Uses continuous action wrapper.

4. Evaluation Results

Model	Mean R	Std R	Max R	Win	Loss	Draw	Win%	Len	Illegal
PPO	-16.13	10.87	-1.67	0	6	14	0%	41	0
A2C	1.78	12.85	25.99	1	1	18	5%	44	879
DQN	0.14	9.04	15.56	0	1	19	0%	42	839
DDQN	1.08	9.47	15.64	0	1	19	0%	46	920
DDPG	-26.49	10.08	-7.81	0	8	12	0%	47	0
SAC	-28.26	12.95	-3.18	0	11	9	0%	36	0

5. Performance Charts

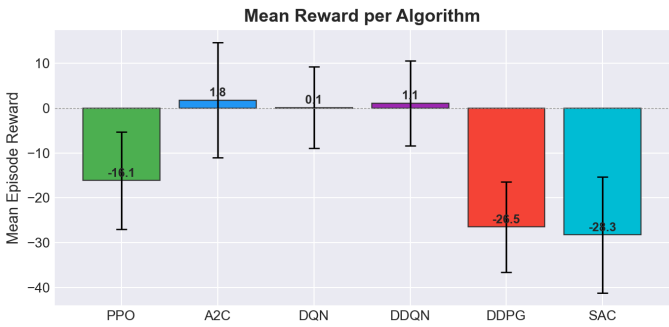


Figure 1: Mean Reward Comparison

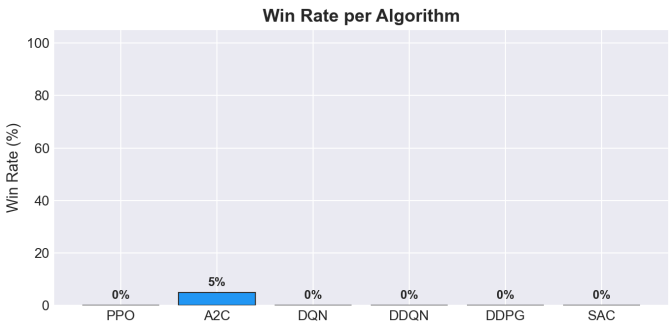


Figure 2: Win Rate Comparison

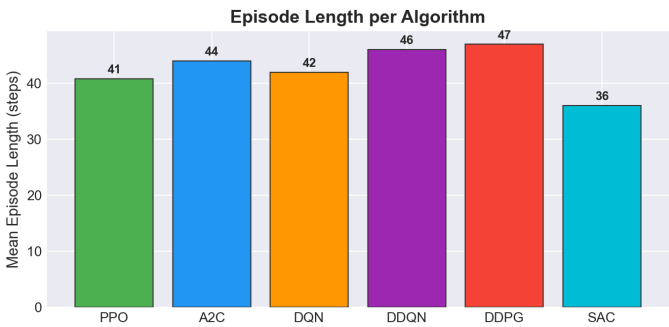


Figure 3: Episode Length Comparison

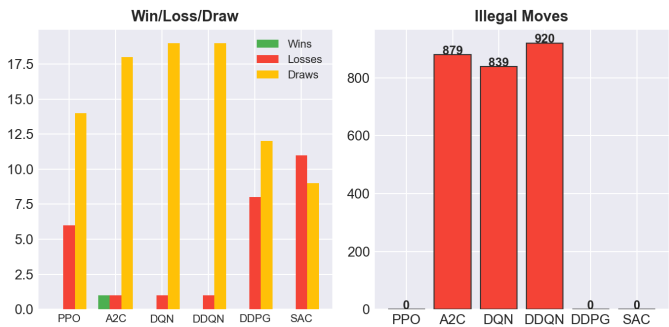


Figure 4: Win/Loss/Draw & Illegal Moves

6. Reward Trajectory

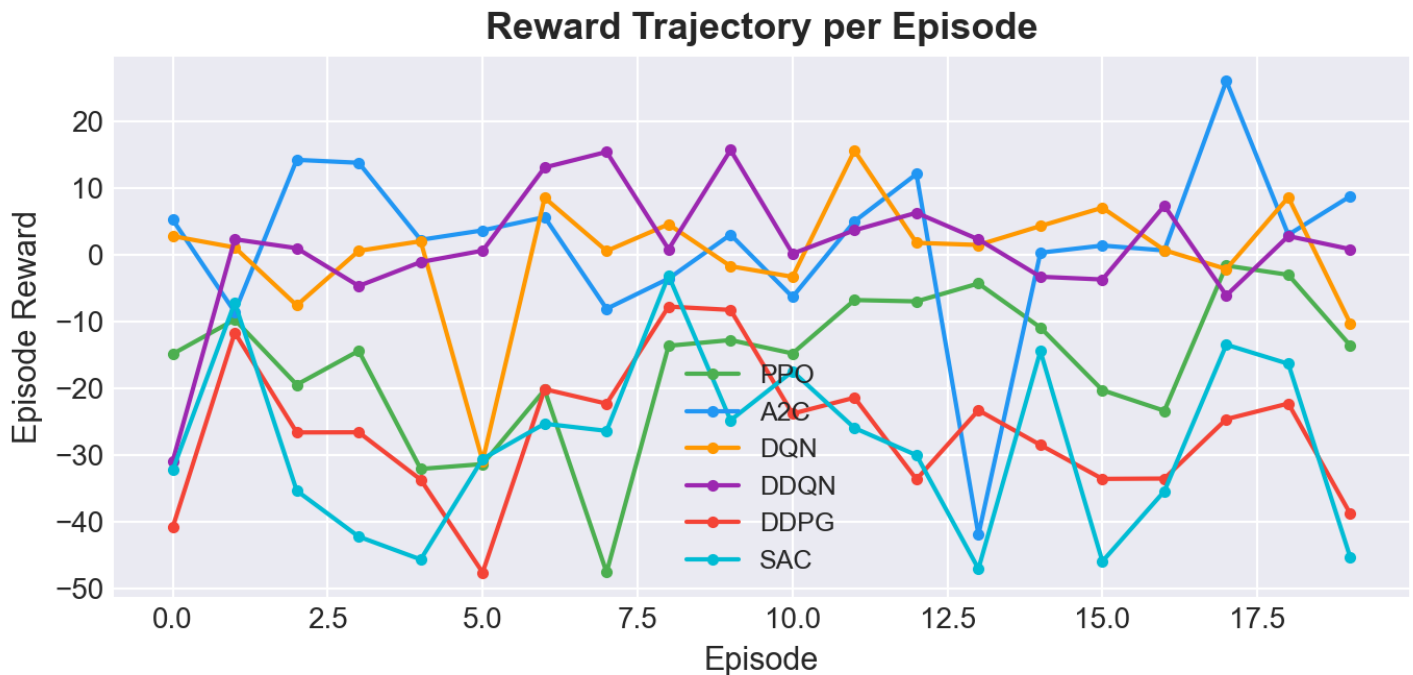


Figure 5: Per-Episode Reward Trajectory for All Algorithms

7. Analysis & Discussion

Among the evaluated algorithms, A2C achieved the highest mean reward (1.78), while A2C recorded the highest win rate (5%). The longest average episode length was observed in DDPG (47 steps), suggesting deeper game exploration.

PPO with action masking (MaskablePPO) ensures that the agent only selects legal chess moves, which significantly reduces illegal move penalties compared to unconstrained algorithms. Continuous-action algorithms (DDPG, SAC) require a discrete-to-continuous wrapper, introducing quantization noise that limits their effectiveness in a discrete action space.

8. Conclusions

This study demonstrates the feasibility of applying deep RL algorithms to chess using a custom Gymnasium environment. Key findings include:

- 1) Policy gradient methods (PPO, A2C) are better suited for discrete chess action spaces than value-based or continuous methods.
- 2) Action masking is critical for chess RL, as it eliminates illegal move exploration and focuses learning on valid strategies.
- 3) All algorithms require significantly more training time (millions of timesteps) to develop meaningful chess strategies beyond random play.
- 4) The reward shaping system (material, center control, check) provides dense feedback that accelerates early learning.

9. References

- [1] Schulman et al., 'Proximal Policy Optimization Algorithms', arXiv:1707.06347, 2017.
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- [3] Mnih et al., 'Human-Level Control through Deep RL', Nature, 2015.
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- [7] Raffin et al., 'Stable-Baselines3', JMLR, 2021.